CISC 473: Deep Learning — Project Proposal

P20: Speech Emotion Recognition with Self-Supervised Pre-Trained Features

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GitHub Repository:
https://github.com/locmai27/CISC-497-speech-emotion-recognition

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Problem Statement and Motivation

Speech emotion recognition (SER) aims to infer a speaker's emotional state from an audio signal. Unlike ordinary speech recognition, the target is not primarily the linguistic content of an utterance, but paralinguistic information such as prosody, pitch, intensity, rhythm, and voice quality. Reliable SER has applications in affective human–computer interaction, conversational agents, accessibility, call-centre analytics, and other systems that need to respond to how something is said rather than only what is said. However, SER remains difficult because labelled emotional-speech datasets are relatively small and emotion is expressed differently across speakers.

Traditional SER systems often rely on handcrafted acoustic features such as mel-frequency cepstral coefficients (MFCCs) followed by recurrent or other supervised classifiers. Recent self-supervised learning (SSL) models such as wav2vec 2.0 and HuBERT instead learn representations from large quantities of unlabelled speech before being adapted to downstream tasks. This offers an attractive approach for SER, where labelled data are limited. At the same time, models pre-trained largely for speech recognition may encode lexical and phonetic information differently from emotion-relevant cues, so it is not obvious which model or Transformer layer provides the most useful representation for emotion recognition.

Our project will compare an MFCC+BiLSTM baseline with wav2vec 2.0 and HuBERT representations on RAVDESS. We will focus on speaker-independent recognition using leave-one-speaker-out (LOSO) cross-validation, so the test speaker is never observed during training. We will also probe individual HuBERT layers and evaluate whether emotion-preserving acoustic augmentation improves generalization across all three approaches. Our main hypothesis is that SSL representations will outperform handcrafted features, but that intermediate representations and targeted augmentation may generalize better to unseen speakers than simply using the final pretrained representation.

Literature Survey

Related Works

Baevski et al. (2020), NeurIPS — wav2vec 2.0. wav2vec 2.0 learns speech representations by masking latent audio representations and solving a contrastive prediction task over quantized targets. The work demonstrated that large-scale self-supervised pre-training can substantially

reduce the amount of labelled speech needed for downstream recognition. Our project uses wav2vec 2.0 as a principal SSL comparison and tests whether representations learned for general speech transfer effectively to emotion classification.

Hsu et al. (2021), IEEE/ACM TASLP — HuBERT. HuBERT uses offline clustering to create hidden-unit targets and trains a BERT-like model to predict masked speech units. It showed that masked prediction over automatically discovered acoustic units can learn strong general-purpose speech representations. HuBERT-base is the primary pretrained model in our project; we will evaluate its emotion-recognition performance and probe each Transformer layer rather than treating the final layer as the only representation.

Yang et al. (2021), Interspeech — SUPERB. SUPERB introduced a common benchmark for evaluating self-supervised speech representations across diverse downstream tasks using lightweight task-specific heads. Its results support the idea that frozen SSL representations can transfer broadly without redesigning the full encoder for every task. Our project follows this representation-probing philosophy, but narrows the analysis to SER and emphasizes speaker-independent evaluation and layer-wise behavior.

Pepino et al. (2021) — Emotion Recognition from Speech Using wav2vec 2.0 Embeddings. Pepino et al. applied pretrained wav2vec 2.0 representations to SER and proposed learning a weighted combination of representations from multiple layers. They evaluated on IEMOCAP and RAVDESS and showed the value of pretrained speech embeddings for emotion recognition. This directly motivates our layer-wise analysis: rather than assuming all layers encode emotion equally, we will measure the predictive value of individual HuBERT layers under a speaker-independent protocol.

Li et al. (2022) — Exploration of a Self-Supervised Speech Model on Emotional Corpora. Li et al. investigated how wav2vec 2.0 representations behave on emotional speech. Their analysis found that intermediate representations can be highly useful for emotion recognition and that the final layer is not necessarily the best layer for paralinguistic information. This finding motivates our systematic HuBERT layer probing and our hypothesis that intermediate HuBERT layers may generalize better to unseen speakers.

Chen et al. (2022), Interspeech — SpeechFormer. SpeechFormer proposes a hierarchical Transformer architecture designed around the temporal structure of speech and evaluates it on tasks including speech emotion recognition. It demonstrates that speech-specific architectural choices can improve efficiency and performance relative to a generic Transformer. While we will not reproduce SpeechFormer, it provides an important SER-focused comparison showing that the organization of speech representations matters, not only model scale.

Ma et al. (2023) — emotion2vec. emotion2vec learns emotion-specific speech representations through self-supervised pretraining. It motivates distinguishing emotion-specific models from the general-purpose HuBERT and wav2vec 2.0 representations examined in our study.

Phukan et al. (2025) — Cross-Corpus SER Benchmarking. This study highlights the importance of paralinguistic pretrained representations for cross-corpus generalization. Our main

evaluation instead focuses on unseen speakers within RAVDESS, using matched conditions to compare representations and augmentation effects.

Open Problem

Previous research shows that pretrained speech features and augmentation can improve emotion recognition. Our project asks whether the same augmentations help MFCC+BiLSTM, HuBERT, and wav2vec 2.0 equally, and which HuBERT layers are most useful for recognizing emotion. We will compare these approaches using the same speaker splits, with each test speaker excluded from training.

Proposed Methodology

Baseline

Our primary dataset will be RAVDESS, containing 1,440 speech audio clips from 24 actors and eight emotion classes. Audio will be resampled and normalized consistently before feature extraction. The classical baseline will extract MFCC sequences with librosa or torchaudio and classify them using a BiLSTM followed by a softmax classification head. This baseline will establish the performance of a conventional handcrafted acoustic representation under the same speaker-independent splits used by the SSL systems.

For pretrained models, we will use publicly available HuBERT-base and wav2vec 2.0 checkpoints through HuggingFace Transformers. We will initially freeze the pretrained encoder and train a lightweight classification head on pooled hidden representations, allowing a controlled comparison of representation quality without requiring expensive end-to-end retraining. If compute and time permit, limited fine-tuning will be evaluated as a secondary experiment.

Proposed Extension

Our first extension is layer-wise HuBERT probing. For every Transformer layer, we will extract its hidden representation, pool it over time, and train the same lightweight classifier. This produces a layer-versus-UAR/WAR curve showing where emotion-relevant information is most accessible. Keeping the probe architecture fixed isolates the contribution of the representation itself.

Our second extension targets speaker-independent generalization. We will evaluate MFCC+BiLSTM, HuBERT features + classifier, and wav2vec 2.0 features + classifier with and without acoustic augmentation. Candidate augmentations are moderate pitch shifting, time stretching, and additive background noise. These transformations are intended to increase acoustic variability without changing the emotion label. We hypothesize that training on these variants will reduce reliance on idiosyncratic speaker cues and improve average LOSO UAR on unseen actors. Common augmentation conditions will be used across all three approaches to compare how their benefits depend on the representation. Augmentation parameters will be fixed before the final evaluation and ablated individually so that any improvement can be attributed to specific transformations. Augmentation will be applied only to training recordings; validation and test recordings will remain unaugmented.

Evaluation Plan

- **Dataset:** RAVDESS as the primary benchmark; CREMA-D may be used as an optional cross-dataset generalization experiment if time permits.

- **Splits:** Leave-one-speaker-out cross-validation across RAVDESS actors, ensuring no test speaker occurs in training.
- **Metrics:** Unweighted average recall (UAR) as the primary metric and weighted average recall (WAR) as a secondary metric; per-emotion confusion matrices will be used for error analysis.
- **Baselines:** MFCC+BiLSTM, wav2vec 2.0 features + classifier, and HuBERT features + classifier.
- **Ablations:** HuBERT layer index; no augmentation vs. pitch shift vs. time stretch vs. noise vs. selected combination for each of the three approaches.
- **Reliability:** Where stochastic training is involved, we will fix random seeds and report mean ± standard deviation over at least three runs, consistent with the capstone reproducibility requirements.

Early Novel Contribution (Optional Bonus)

We propose a controlled study of speaker-independent HuBERT layer selection combined with emotion-preserving acoustic augmentation. Rather than using the final HuBERT layer by default, we will select the layer using validation data within each LOSO training fold and test whether targeted acoustic augmentation improves its generalization to the held-out speaker. The contribution is therefore not a new large architecture, but a reproducible analysis connecting representation depth, speaker independence, and augmentation. A positive result would provide a lightweight recipe for improving SER without training a speech encoder from scratch; a negative result would still characterize when these commonly used augmentations fail to improve speaker-independent transfer.

Project Timeline and Roles

Responsibilities are divided by modelling path. Each member owns the literature review, implementation, tuning, experiments, analysis, and writing for their respective models.

- **Anastasia — Project Lead / Research & Engineering Lead (MFCC + BiLSTM):** Coordinates milestones and submissions; implements shared data preprocessing and augmentation utilities; manages the MFCC + BiLSTM modelling path.
- **Vu — Research & Engineering Lead (HuBERT & wav2vec 2.0):** Implements shared experiment infrastructure, evaluation, and logging; manages the HuBERT and wav2vec 2.0 modelling paths, including HuBERT layer-wise probing.
- **Both:** Agree on research questions and experimental design; synthesize related work; compare findings across models; jointly prepare the report and presentation and verify reproducibility.

Timeline

Shared setup and model development may overlap once interfaces are agreed. Both paths use identical speaker splits, metrics, and common augmentation conditions. Model and augmentation settings are selected using validation data, without using held-out test speakers.

Weeks	Task Group and Activities	Responsible
1–2	**Research and planning.** Review literature for each modelling path; jointly define hypotheses, experiment scope, and evaluation protocol; finalize the proposal. **Proposal due: Week 2.**	Each: own literature review. Both: study design and proposal.
1–3	**Shared infrastructure.** Set up repository/environment, data loading interfaces, speaker split generation, evaluation metrics, logging, and common result formats.	Vu; both agree on interfaces and evaluation rules.
1–3	**Shared preprocessing.** Prepare RAVDESS, verify labels and speaker IDs, standardize audio, and implement reusable augmentation functions.	Anastasia; both agree on preprocessing and augmentation settings.
3–4	**Initial model implementation.** Build and validate each modelling path; obtain initial results without augmentation. Initially freeze SSL encoders and train their classifiers.	Vu: HuBERT/wav2vec 2.0. Anastasia: MFCC + BiLSTM.
4–6	**Model experiments.** Tune each path using validation data; run HuBERT layer-wise probes; begin matched augmentation experiments; consolidate preliminary findings. **Midterm due: Week 6.**	Each: own model experiments. Vu: layer probing. Both: midterm.
6–8	**Full evaluation and individual analysis.** Complete agreed augmentation ablations and LOSO evaluation; repeat stochastic experiments with at least three independent seeds. Produce each path's results, plots, confusion matrices, and failure analysis.	Each: own modelling path.
8–9	**Comparison and report writing.** Write model-specific methods and results; compare model performance and augmentation effects; jointly develop the discussion, limitations, and conclusions.	Each: own sections. Both: comparative analysis and report integration.
10	**Reproducibility and delivery.** Run each other's code; finalize README, report, code release, and presentation. **Final due: Week 10.**	Both.

Contribution Declaration

Member	Primary Contributions	Estimated
Tumanov, Anastasia	Project coordination and submissions; shared dataset preparation, audio preprocessing, and augmentation utilities; research, implementation, experiments, analysis, and writing for MFCC + BiLSTM; joint comparative analysis and final deliverables.	50%
Mai, Vu Thanh Loc	Shared experiment infrastructure, speaker splits, evaluation, and logging; research, implementation, experiments, analysis, and writing for HuBERT/wav2vec 2.0; HuBERT layer probing; joint comparative analysis and final deliverables.	50%

Contribution percentages are provisional and will be updated to reflect actual work. Both members conduct research and engineering within their modelling paths and review each other's work.

References

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